

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	A,C	C	A,B,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.16 to 0.17	1.00	1.51	0.50	1.41	4.50	0.50	0.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,C,D	A,C,D	A,B	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	23.30	1.50	6.50	2.00	78.00	49.25	6.90 to 6.92	0.57

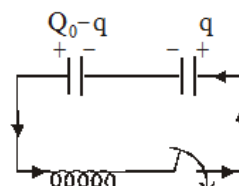
PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B	A,C	A,C	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	5.00	5.00	4.00	0.00	0.00	5.00	0.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (A,C)**

$$-L \frac{di}{dt} - \frac{q}{c} + \frac{(Q_0 - q)}{c} = 0$$



$$\frac{-Ld^2q}{dt^2} - \frac{2q}{c} + \frac{Q_0}{c} = 0$$

$$\frac{d^2q}{dt^2} = -\frac{2}{LC} \left[q - \frac{Q_0}{2} \right]$$

$$\text{frequency} = \frac{1}{\pi\sqrt{2LC}}$$

$$\text{max current} = \frac{Q_0}{2}\omega = \frac{d_0}{2} \sqrt{\frac{2}{LC}} = \frac{Q_0}{2LC}$$

 2. **Ans (A,C)**

$$\text{Current in secondary circuit } i = \frac{3}{R+2R} = \frac{3}{3R} = \frac{1}{R}$$

$$\text{current through } 3R = \frac{2 \times (1/R)}{3} = \frac{2}{3R}$$

$$\text{P.d. across } xy = \frac{2}{3R} \times 3R = 2 \text{ volt}$$

$$\text{P.d. across A and C} = \frac{25}{100} \left[\frac{10}{r+8} \right] \times 8$$

$$2 = \frac{1}{4} \left[\frac{10}{r+8} \right] \times 8 \quad 3r + 32 = 40 \quad r = 2\Omega$$

 3. **Ans (A,C)**

$$\text{So } t = \frac{2\ell + \pi R}{V_0}$$

4. **Ans (C)**

$$Z_{RL} = \sqrt{R^2 + \omega^2 L^2}$$

$$Z_{RLC} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\cos \phi_{RL} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\cos \phi_{RLC} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

According to question

$$(\omega L)^2 = \left(\omega L - \frac{1}{C\omega}\right)^2 \Rightarrow C = \frac{1}{2\omega^2 L}$$

5. **Ans (A,B,D)**

From 2nd graph, when $\theta = \frac{\pi}{2}$, $\phi = 4\pi = 2\lambda$
= distance between S_1 & S_2

At $\theta = \frac{\pi}{6}$, $\phi = d \sin \theta = 2\lambda \frac{\ell}{2} = \lambda$.

6. **Ans (A,B,C)**

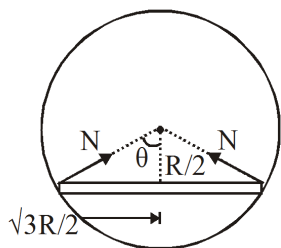
$$\frac{1}{2} \left[\frac{M(\sqrt{3}R)^2}{12} + M \left[\frac{R}{2} \right]^2 \right] \omega^2 = mg \frac{R}{2}$$

$$\Rightarrow \frac{1}{2} \left[\frac{MR^2}{4} + \frac{MR^2}{4} \right] \omega^2 = mg \frac{R}{2}$$

$$\Rightarrow \frac{MR^2}{2} \omega^2 = MgR$$

$$\Rightarrow \omega = \sqrt{\frac{2g}{R}}$$

$$V_{cm} = \omega \frac{R}{2} = \sqrt{\frac{2g}{R}} \times \frac{R}{2} = \sqrt{\frac{9R}{2}}$$



$$2N \cos \theta = M \omega^2 \frac{R}{2}$$

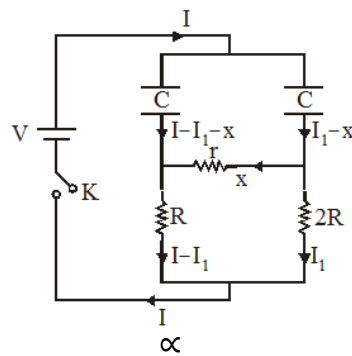
$$2N \times \frac{R/2}{R} = M \times \frac{2g}{R} \times \frac{R}{2}$$

$$N = Mg$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**



Our target is $\int_0^\infty x dt$

$$\int_0^\infty I dt = 2 CV \text{ Now Let } \int_0^\infty x dt = q \text{ and } \int_0^\infty I_1 dt = q_1$$

Now applying loop rule $2I_1 R = rx + R(I - I_1)$

$$\Rightarrow 3I_1 = \frac{r}{R} x + I$$

$$3 \int_0^\infty I_1 dt = \frac{r}{R} \int_0^\infty x dt + \int_0^\infty I dt$$

$$\Rightarrow 3 q_1 = \frac{r}{R} q + 2CV \quad (i)$$

$$\text{Also } \int_0^\infty (I_1 + x) dt = CV$$

$$\therefore q_1 + q = CV \quad (ii)$$

$$\text{Solving (1) and (2) } q = \frac{CV}{3 + r/R}$$

8. **Ans (C)**

For (A) : $2A = 0.06 \Rightarrow$ Amplitude of constituent wave $A = 0.03$ m

For (B) : Position of node when $y = 0$

$$\Rightarrow \sin(2\pi x) = 0 \Rightarrow 2\pi x = n\pi \text{ where } n=1,2,3\ldots$$

$$\Rightarrow x = \frac{n}{2} = \frac{1}{2} = 0.5 \text{ m}$$

For (C) : Position of antinode $0.06 = 0.06 \sin(2\pi x)$

$$\Rightarrow 2\pi x = (2n-1)\frac{\pi}{2} \text{ where } n=1,2,3\ldots$$

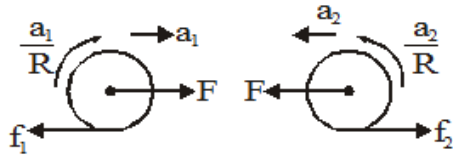
$$\Rightarrow x = \frac{1}{4} = 0.25 \text{ m}$$

For (D) : Amplitude at $x = \frac{1}{12}$ m

$$\Rightarrow A = 0.06 \sin\left(2\pi \times \frac{1}{12}\right) = 0.06 \sin \frac{\pi}{6}$$

$$= 0.06 \times \frac{1}{2} = 0.03 \text{ m}$$

9. Ans (A)



Let v_1 = Velocity of A and v_2 = velocity of B

just before they strike by energy conservation

$$0 = \frac{k(20)(-0)}{2R} + \frac{1}{2}(2m)v_1^2 \left[1 + \frac{2}{5} \right]$$

$$+ \frac{1}{2}(m)v_2^2 \left[1 + \frac{2}{3} \right]$$

$$\Rightarrow \frac{7}{5}mv_1^2 + \frac{5}{6}mv_2^2 = \frac{kQ^2}{R}$$

By taking torque about contact points

$$FR = \left[\frac{7}{5}(2m)R^2 \right] \frac{a_1}{R} \text{ \& } FR = \left[\frac{5}{3}(m)R^2 \right] \frac{a_2}{R}$$

$$\Rightarrow 42a_1 = 25a_2 \Rightarrow 42v_1 = 25v_2$$

$$\Rightarrow \frac{7}{5}mv_1^2 + \frac{5}{6} \left(\frac{42}{25} \right)^2 v_1^2 = \frac{kQ^2}{R}$$

$$\Rightarrow \frac{469}{125}mv_1^2 = \frac{kQ^2}{R} \Rightarrow mv_1^2 = \frac{125kQ^2}{469R}$$

Therefore

$$\text{For (A) KE of system} = \frac{kQ^2}{R}$$

$$\text{For (B) KE of A} = \frac{7}{5}mv_1^2 = \frac{25kQ^2}{67R}$$

$$\text{For (C) KE of B} = \frac{5}{6}mv_2^2 = \frac{42kQ^2}{67R}$$

For (D) Rotational KE of system

$$= \frac{2}{5}mv_1^2 + \frac{1}{3}mv_2^2 = \frac{838kQ^2}{2345R}$$

10. Ans (D)

For A :

$$B_{P_1} = 4 \times \frac{\mu_0 I}{4\pi(L/2)} \left[\sin \frac{\pi}{4} + \sin \frac{\pi}{4} \right] = \frac{2\sqrt{2}\mu_0 I}{\pi L}$$

For B :

$$B_{P_2} = \frac{2kM}{r^3} = \frac{2 \left(\frac{\mu_0}{4\pi} \right) (IL^2)}{a^3} = \frac{\mu_0 IL^2}{2\pi a^3}$$

For C :

$$B_{P_3} = \frac{kM}{r^3} = \frac{\left(\frac{\mu_0}{4\pi} \right) (IL^2)}{a^3} = \frac{\mu_0 IL^2}{4\pi a^3}$$

For D :

$$B_{P_4} = \frac{kM}{r^3} \sqrt{1 + 3\cos^2\theta} \text{ where } \cos\theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow B_{P_4} = \frac{\left(\frac{\mu_0}{4\pi} \right) (IL^2)}{(a\sqrt{2})^3} \sqrt{1 + 3 \left(\frac{1}{2} \right)} = \frac{\mu_0 \sqrt{5} IL^2}{16\pi a^3}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.16 to 0.17)

$$F = \sigma \phi$$

$$= \frac{Q}{a^2} \times \frac{q}{6\epsilon_0}$$

2. Ans (1.00)

$$v^2 = I \Rightarrow (v_0 - Ir)^2 = I$$

3. Ans (1.51)

$$\cos A = \frac{1}{8} \Rightarrow 1 - 2\sin^2 \frac{A}{2} = \frac{1}{8}$$

$$\Rightarrow \sin \frac{A}{2} = \frac{\sqrt{7}}{4}$$

$$\text{now, required } \mu = \cot \sec \frac{A}{2} = \frac{4}{\sqrt{7}}$$

4. Ans (0.50)

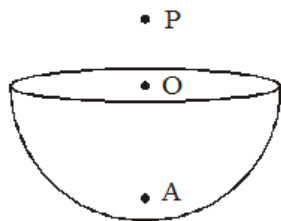
$$Q = dU + W$$

$$= \frac{nR\Delta T}{\gamma - 1} + \frac{nR\Delta T}{1 - \eta}$$

$$= 2RT_0$$

So the value of x is 2.

5. Ans (1.41)



At point p potential is $V_P = \frac{-2GM}{R} + \frac{\sqrt{2}GM}{R}$

$$V_0 = -\frac{GM}{R}, V_A = \frac{-\sqrt{2}GM}{R}$$

$$\text{Now } V_P - V_0 = \frac{GM}{R} + \frac{\sqrt{2}GM}{R} \text{ and } \frac{-GM}{R} + \frac{\sqrt{2}GM}{R}$$

Now applying conservation of energy

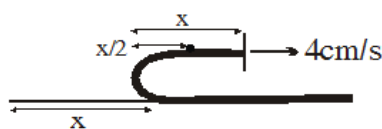
$$\frac{1}{2}mv_0^2 = \left[\frac{-GM}{R} + \frac{\sqrt{2}GM}{R} \right]$$

Again

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + m \left[\frac{-GM}{R} + \frac{\sqrt{2}GM}{R} \right]$$

$$\Rightarrow V = \sqrt{2}V_0$$

6. Ans (4.50)



$$\frac{d(2x)}{dt} = 4\text{m/s}; \frac{2dx}{dt} = 4; \frac{dx}{dt} = 3 \text{ cm/s}$$

$$\frac{d}{dt} \left(x + \frac{x}{2} \right) \Rightarrow \frac{3dx}{2dt} = \frac{3}{2} \times 3 = \frac{9}{2} \text{ cm/sec}$$

7. Ans (0.50)

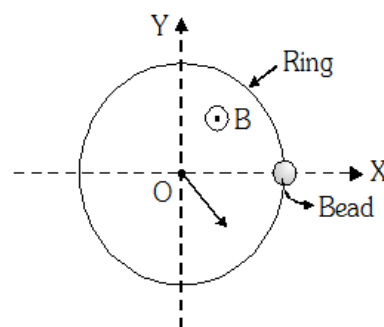
$$E_2 = M \frac{di_1}{dt} = 2ktM$$

$$i_2 = \frac{E_2}{R} = \frac{2ktM}{R}$$

$$\int dq = \int i_2 dt = \frac{2Mk}{R} \int_0^T t dt = \frac{kMT^2}{R}$$

$$\therefore \frac{\lambda}{1-\lambda} = 1 \Rightarrow \lambda = \frac{1}{2} \Rightarrow \frac{1}{\lambda} = 2.$$

8. Ans (0.00)



B depends upon r.

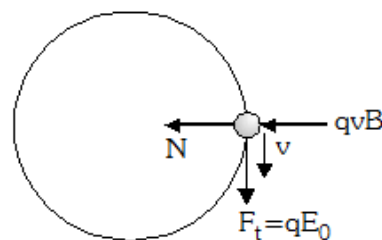
$\therefore$ Flux linked with ring :

$$\Phi = \int_0^R B 2\pi r dr = \int_0^R \frac{E_0 t}{r} 2\pi r dr$$

$$\Phi = 2\pi E_0 t R$$

$$\frac{d\Phi}{dt} = 2\pi E_0 R = (\text{induced electric field}) 2\pi R$$

$\therefore$ Induced electric field = E_0 (tangential)



Circular motion of bead :

$$F_t = qE_0 = \frac{mdv}{dt} \Rightarrow v = \frac{qE_0 t}{m}$$

Centripetal force equation :

$$N + qvB = \frac{mv^2}{R}$$

$$N + q \frac{qE_0 t}{m} \frac{E_0 t}{R} = \frac{m}{R} \frac{q^2 E_0^2 t^2}{m^2}$$

$$N = 0$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (A,C,D)

Difference in refractive indices of DM and DP cause easy detection by ultramicroscope.

Less number of particles gives less (negligible) colligative properties.

4. Ans (A,C,D)

$$pH = 3 \Rightarrow [H^+] = 10^{-3}M$$

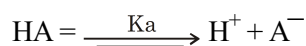
let the degree of dissociation be α

$$\Rightarrow \alpha = \frac{10^{-3}}{0.1} = 0.01 \Rightarrow \% \alpha = 1$$

$$\Rightarrow i = 1 + (y - 1)\alpha = 1 + (2 - 1)(0.01) = 1.01$$

$$\Rightarrow \pi(\text{osmotic pressure}) = (1.01)(0.1)RT = 0.101 RT$$

$$i = \frac{M_{th}}{M_{ob}} \Rightarrow M_{th} = 1.01 \times M_{ob}$$



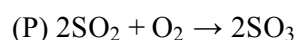
$$K_a = \frac{C\alpha^2}{1 - \alpha} \cong 10^{-1} \times 10^{-2} = 10^{-3}M$$

$$K_b(A^-) = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-3}} = 10^{-11}M \text{ at } 25^\circ C$$

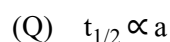
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A)



$$\frac{-d[SO_2]}{dt} = \frac{+d[SO_3]}{dt} \Rightarrow \frac{d(\text{wt.}SO_3)}{dt} = 1.2$$



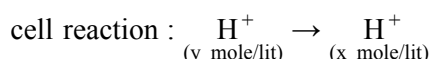
$$\Rightarrow \text{order of reaction} = 0$$

$$(R) t_{1/2} = \frac{0.693}{k} = 1 \text{ min}$$

$$\Rightarrow \frac{0.025}{0.2} = \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

$$\Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^{t/1}$$

$$\Rightarrow t = 3 \text{ min}$$



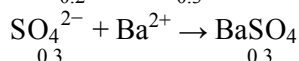
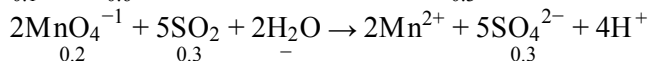
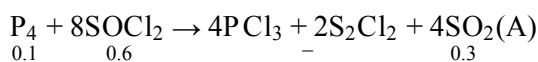
$$E = 0.0177 = \frac{0.059}{1} \log \frac{y}{x}$$

$$\Rightarrow \log \frac{y}{x} = 0.3 \Rightarrow \frac{y}{x} = 2$$

PART-2 : CHEMISTRY

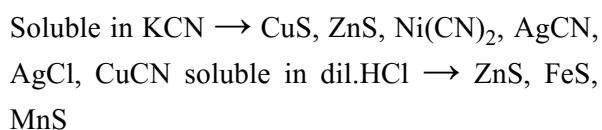
SECTION-II

1. Ans (23.30)

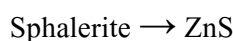
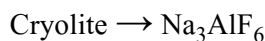
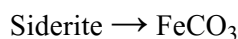
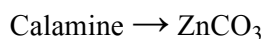


$$\Rightarrow x = 0.3 \times [137 + 32 + 64] = 69.9g$$

2. Ans (1.50)

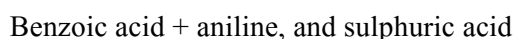


3. Ans (6.50)



$$\Rightarrow 13 \text{ types of elements}$$

4. Ans (2.00)



5. Ans (78.00)

$$\text{Molarity of 1}^{\text{st}} \text{ solution} = \frac{0.2 \times M}{\frac{0.2M + 18 \times 0.8}{d_1}} \times 1000$$

$$\text{Molarity of 2}^{\text{nd}} \text{ solution} = \frac{0.1}{\frac{0.1M + 0.9 \times 18}{d_2}} \times 1000$$

$$\Rightarrow d_2 = 0.8d_1$$

$$\frac{1}{2} \times \frac{0.2 \times 1000}{0.2M + 18 \times 0.8} \times d_1 = \frac{0.1 \times 0.8d_1}{0.1M + 0.9 \times 18}$$

$$\Rightarrow \frac{0.2 \times 1000}{0.2M + 18 \times 0.8} = \frac{0.1 \times 0.8}{0.1M + 0.9 \times 18}$$

$$M = 78$$

8. Ans (0.57)

$$(X = 4, Y = 7)$$

$$X \div Y = 0.57$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B)

Each ring has 15 positions total attempt = 15^3

but only are correct attempt.

$N = 15^3 - 1 = 2 \times 7 \times 241$ clearly divisible by 482

2. Ans (A,B)

The child standing at even numbered position can not be the first child to face the problem of change.

3. Ans (A,C)

$\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$ coplanar with $\vec{b}, \vec{c}$

$$\Rightarrow -3x + y - z = 0$$

projection of $\vec{d}$ on $\vec{a} = \frac{|\vec{a} \cdot \vec{d}|}{|\vec{a}|}$ so

$$\sqrt{\frac{2}{3}} = \frac{1}{\sqrt{6}} |2x - y + z|$$

$$2x - y + z = \pm 2$$

4. Ans (A,C)

$$|z - z_1| + |z - z_2| = \lambda, \text{ locus is real if } f\lambda > |z_1 - z_2|$$

5. Ans (A,B,C,D)

$$(2\lambda - 2)2 - 2(1 - 2\lambda) - (3 - \lambda) = 0 \Rightarrow \lambda = 1$$

$$P(1, 1, -1)$$

A point on L at distance r from (x_1, y_1, z_1) is

$$(x_1 + \ell r, y_1 + mr, z_1 + nr)$$

$$\ell = 2/3, m = -2/3, n = -1/3, r = \pm 3$$

$$Q(x = 3, y = -1, z = -2) \equiv (x_1, y_1, z_1)$$

$$R(x = -1, y = 3, z = 0) \equiv (x_2, y_2, z_2)$$

$$\text{Normal vector } \vec{n} = \vec{n}_1 \times \vec{n}_2 = \hat{i} + 15\hat{j} + 16\hat{k}$$

$$\text{Equation of planes } 1(x - 3) + 15(y + 1) + 16(z + 2)$$

$$= 0$$

$$\text{and } 1(x + 1) + 15(y - 3) + 16(z) = 0$$

6. Ans (A,B,C,D)

$$(A) \sqrt{x} + \sqrt{y} = 1 \Rightarrow x^2 - 2xy + y^2 - 2x - 2y + 1 = \text{parabola}$$

$$(B) \frac{1}{x} + \frac{1}{y} = 1 \text{ hyperbola}$$

$$(C) \frac{1}{x} + \frac{1}{y} + \frac{1}{x+y} = 0 \Rightarrow x^2 + 3xy + y^2 = 0 \text{ pair of lines}$$

$$(D) x + y - \frac{1}{x+y} = 0 \Rightarrow x + y = \pm 1 \text{ pair of lines}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

(A) using 2nd definition of ellipse

$$(B) \text{ Let } \alpha = 3h - 2, \beta = 3k \Rightarrow \frac{\alpha + 2}{3} = h, \frac{\beta}{3} = k$$

(C) 2nd definition of hyperbola

(D) circle with given diagonal as diameter

8. Ans (D)

$$(A) a_i \in P, a_i \in Q$$

$$(B) a_i \in P, a_i \notin Q$$

$$(C) a_i \notin P, a_i \in Q$$

$$(D) a_i \notin P, a_i \notin Q$$

For $P \cup Q = A$ or $P \cap Q = \phi$, no of ways = 3^n

singleton set in nC_1 ways. Rest of the elements in 3^{n-1} ways $P \cup Q = \phi$ in just one way.

10. Ans (B)

$$A \rightarrow R; B \rightarrow P, R; C \rightarrow Q, R, S; D \rightarrow P, Q, R$$

$$(A) \sin^2 2x + \cos^4 2x + 2 = \frac{39}{16} + \frac{1}{16} \cos 8x$$

$$(B) (2^x - 2)^2 + 1 + |\lambda - 1| - 3 = |\sin y| \Rightarrow \lambda - 1 = \pm 3$$

$$(C) f(x) < a \text{ for at least one real } x, x = 2, 4, 6.$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3.00)

$$\tan \theta = t$$

$$\int e^t \frac{(1-t)^2}{(1+t^2)} dt = \frac{e^t}{1+t^2} + K, A + B = 3$$

2. **Ans (5.00)**

$$\begin{aligned}
 &= \lim_{h \rightarrow \infty} \frac{1}{5^{n-1}} \sum_{r=1}^n {}^nC_r \left(\sum_{i=0}^{r-1} {}^rC_i 3^i \right) \\
 &= \lim_{h \rightarrow \infty} \frac{1}{5^{n-1}} \sum_{r=1}^n {}^nC_r (4^r - 3^r) \\
 &= \lim_{n \rightarrow \infty} \frac{5^n - 4^n}{5^{n-1}} = 5
 \end{aligned}$$

 3. **Ans (5.00)**

$$\begin{aligned}
 z_1^3 - 3z_1z_2^2 + 3iz_1^2z_2 - iz_2^3 &= 2 + 11i \\
 |z_1^2 + z_2^2| &= |(z_1 + iz_2)(z_1 - iz_2)| = 5
 \end{aligned}$$

 4. **Ans (4.00)**

Conceptual

 6. **Ans (0.00)**

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{bmatrix} \Rightarrow |BA| = 0$$

 7. **Ans (5.00)**

$$\begin{aligned}
 y^2 &= P(x), 2y \frac{dy}{dx} = P'(x) \\
 \Rightarrow 2y \frac{dy}{dx^2} + 2y_1^2 &= P''(x) \Rightarrow 2y^3y_2 + 2y^2y_1^2 = y^2P''(x)
 \end{aligned}$$

 8. **Ans (0.00)**

$$\begin{aligned}
 [\sqrt{2046}] &= [\sqrt{2047}] = [\sqrt{2048}] = [\sqrt{2049}] = 45 \\
 2003^{\text{rd}} \text{ term} &= 2003 + 45 = 2048
 \end{aligned}$$